SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Muthu Krishnan K / 192511039_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Muthu Krishnan K

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______→ Transport → Network → ______→ Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:
DataLink→MACAddress→LocalDelivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:
ErrorControl
→DataLinkLayer→ErrorDetection
→ Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:
Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖
Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical

Analyze the missing layers and explain how their functions are essential for reliable communication.

Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Missing Layers:

1. **Session Layer**
2. **Data Link Layer**

Explanation:

Session Layer

- Establishes, manages, and terminates communication sessions between devices.
- Maintains synchronization and checkpoints during data transfer.
- Helps recover communication after interruptions.

Data Link Layer

- Provides framing of data.
- Uses MAC addresses for local delivery.
- Detects transmission errors and controls access to the communication medium.

Answer:

The missing layers are **Session Layer** and **Data Link Layer**. Together they ensure reliable communication through session management, framing, and error detection.

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2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

Concept Map:

Transport Layer → Segment
↓
Network Layer → Packet
↓
Data Link Layer → Frame
↓
Physical Layer → Bits

Explanation:

OSI Layer	Data Unit
Transport Layer	Segment
Network Layer	Packet
Data Link Layer	Frame
Physical Layer	Bits

Analysis:

1. The Transport Layer divides data into **segments**.
2. The Network Layer adds IP addressing to create **packets**.

3. The Data Link Layer adds MAC addressing and error-checking information to create **frames**.
4. The Physical Layer converts frames into **bits** for transmission through cables or wireless media.

Answer:
The transformation from Segment → Packet → Frame → Bits enables efficient addressing, routing, error detection, and physical transmission of data.

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3. Given the concept map:

Data Link → MAC Address → Local Delivery
Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.
Concept Map:
Data Link Layer
↓
MAC Address
↓
Local Network Delivery
Network Layer
↓
IP Address
↓
Inter-Network (Global) Delivery
Explanation:
MAC Address

- Used within the local network.
- Helps switches identify destination devices.
- Ensures delivery inside a LAN.

IP Address

- Used across different networks.
- Helps routers determine the path to the destination.
- Enables communication over the Internet.

Answer:
MAC addresses provide local delivery, while IP addresses provide global routing. Together they ensure complete end-to-end data delivery across networks.

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4. A concept map shows:

Error Control
↓
Data Link Layer → Error Detection
Transport Layer → Error Correction
Analyze how the interaction between these layers improves reliability in data transmission.
Concept Map:
Error Control
↓
Data Link Layer
↓
Error Detection
↓

Transport Layer
↓
Error Correction

Explanation:

Data Link Layer

- Detects errors using CRC and checksums.
- Identifies corrupted frames during transmission.

Transport Layer

- Uses acknowledgments and retransmissions.
- Corrects errors by requesting lost or damaged data again.
- Ensures data arrives in the correct sequence.

Analysis:

- Error Detection identifies transmission problems.
- Error Correction recovers missing or corrupted data.
- Together they improve communication reliability.

Answer:

The Data Link Layer detects errors while the Transport Layer corrects them through retransmission and sequencing, ensuring reliable communication.

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5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

Concept Map:

Application Layer
↓ HTTP Request
Presentation Layer
↓ Encryption (HTTPS)
Session Layer
↓ Session Establishment
Transport Layer
↓ TCP Connection
Network Layer
↓ IP Routing
Data Link Layer
↓ Frame Transmission
Physical Layer
↓ Signal Transmission

Analysis:

Failed Layer	Impact
Application	Website cannot be accessed
Presentation	Encryption/compression fails
Session	Connection cannot be maintained
Transport	Data loss and retransmission issues
Network	Routing failure
Data Link	Frame delivery failure
Physical	No signal transmission

Since every layer depends on the layer below it, failure in one layer affects all higher layers.

Answer:

Web browsing requires all OSI layers to work together. Failure of any layer interrupts communication and prevents successful webpage access.

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6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✖ → Transport ✖ → Application ✖

Analyze the problem and explain how the concept map helps in troubleshooting.

Concept Map:

Physical Layer ✔
↓
Data Link Layer ✔
↓
Network Layer ✖
↓
Transport Layer ✖
↓
Application Layer ✖

Analysis:

Physical Layer Working

- Cables, wireless signals, and hardware are functioning.

Data Link Layer Working

- Frames are being transmitted successfully.

Network Layer Failed

- IP addressing or routing problem exists.
- Router configuration may be incorrect.
- IP address may be missing or invalid.

Since the Network Layer fails:

- Transport Layer cannot establish connections.
- Applications cannot communicate.
- Services such as web browsing, email, and FTP stop working.

Troubleshooting Steps:

1. Check IP address configuration.
2. Verify subnet mask and gateway.
3. Test router connectivity.
4. Use ping and traceroute commands.
5. Check routing tables.

Answer:

The failure occurs at the **Network Layer**. The concept map helps identify the exact layer causing the issue, making troubleshooting faster and more systematic.